

Limits of Schur Rings

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A Bit of Notation

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The Frobenius Map

If $C \subseteq G$ and $m \in \mathbb{Z}$, then

$$C^{(m)} = \{g^m \mid g \in C\} \subseteq G$$

and

$$C^* = \{g^{-1} \mid g \in C\} \subseteq G.$$

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Simple Quantities

If $C \subseteq G$, then

$$\overline{C} = \sum_{g \in C} g \in F[G].$$

Schur Rings

Definition - Schur Rings

Let $\mathcal{P}$ be a partition of finite support for the group G and let $\mathfrak{S} = \text{Span}\{\overline{C} \mid C \in \mathcal{P}\}$. We say $\mathfrak{S}$ is a **Schur Ring** (or S-ring) if

- (i) $\{1\} \in \mathcal{P}$,
- (ii) $C^* \in \mathcal{P}$ whenever $C \in \mathcal{P}$,
- (iii) $\overline{C} \overline{D} \in \mathfrak{S}$, whenever $C, D \in \mathcal{P}$.

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Schur Rings

The elements of a Schur partition $\mathcal{P}$ are called the **primitive sets** of S .

Any subset of G which is a union of primitive sets of S is called an **S -set**.

A subgroup $H \leq G$ which is also an S -set is called an **S -subgroup**.

For an S -subgroup H , all the primitive sets contained in H form a Schur partition of H . The associated Schur ring over H is called a **Schur subring** of S .

Schur Rings

Schur rings were first studied by Schur and Wielandt in the early 20th century, but their focus (and the focus of others) was always on finite groups.

Work by Bastian, Brewer, Humphries, M, and Thompson extended Schur rings to any group, including infinite groups.

Association Schemes

In the 1970's and 1980's, connections between Schur rings and algebraic combinatorics were beginning to be found.

In 1985, Klin showed that Schur rings can be used to form association schemes:

The group ring $\mathbb{Z}[G]$ coincides to the association scheme $\mathcal{A} = \{R_g \mid g \in G\}$ on the set G by the relations

$$R_g = \{(h, k) \mid h = gk\}.$$

Association Schemes

More generally, a Schur ring $S \subseteq \mathbb{Z}[G]$ coincides to the association scheme $\mathcal{A} = \left\{ \bigcup_{g \in C} R_g \mid C \in \mathcal{P} \right\}$, where $\mathcal{P}$ is the partition associated to S .

Due to this connection, discoveries about Schur rings can often be reinterpreted as discoveries about association schemes.

For example, Leung and Man (1996, 1998) introduced the wedge product of Schur rings to classify Schur rings over cyclic groups.

Independently, Evdokimov and Ponomarenko (2001, 2002) presented the same construction called the “generalized wreath product.”

Association Schemes

Muzychuk (2009) generalizes this construction to association schemes.

Hendrickson (2012) connects Schur rings with supercharacter theories defined by Diaconis and Issacs (2008). He then generalizes the wedge product to supercharacter theories.

The extension of Schur rings to infinite groups may prove to be the trailblazer for association schemes' and supercharacter theories' path to the infinite.

Examples of Schur Rings

Examples of Schur Rings:

- i) Every group algebra $F[G]$ is a Schur ring with partition $G = \bigcup_{g \in G} \{g\}$, called the **discrete Schur ring**.

e.g. $G = \mathbb{Z}_6 = \langle a \rangle$:

$$S = \text{Span}\{1, a, a^2, a^3, a^4, a^5\}$$

Examples of Schur Rings

Examples of Schur Rings:

- ii) Let $\mathcal{P} = \{\{1\}, G \setminus \{1\}\}$. This Schur ring is known as the **Trivial Schur ring**.

e.g. $G = \mathbb{Z}_6 = \langle a \rangle$:

$$S = \text{Span}\{ \textcolor{red}{1}, \textcolor{green}{a} + \textcolor{green}{a}^2 + \textcolor{green}{a}^3 + \textcolor{green}{a}^4 + \textcolor{green}{a}^5 \}$$

Examples of Schur Rings

Examples of Schur Rings:

- iii) Let $\mathcal{H} \leq \text{Aut}(G)$ and let S be the Schur ring afforded by the automorphism classes of $\mathcal{H}$, called an **orbit Schur ring**.

e.g. $G = \mathbb{Z}_7 = \langle a \rangle$, $\mathcal{H} = \langle 2 \rangle$:

$$S = \text{Span}\{ \textcolor{red}{1}, \textcolor{green}{a} + \textcolor{green}{a}^2 + \textcolor{green}{a}^4, \textcolor{blue}{a}^3 + \textcolor{blue}{a}^5 + \textcolor{blue}{a}^6 \}$$

Examples of Schur Rings

Examples of Schur Rings:

- iv) Let $H \times K = G$. Let $\mathfrak{S}$ and $\mathfrak{T}$ be Schur rings over H and K , respectively. Then $\mathfrak{S} \otimes_F \mathfrak{T}$ is a Schur ring over G , called the **Direct Product** of $\mathfrak{S}$ and $\mathfrak{T}$. Its partition is constructed by all the possible products of sets from the partitions of $\mathfrak{S}$ and $\mathfrak{T}$.

e.g. $G = \mathbb{Z}_6 = \langle a \rangle$, $H = \mathbb{Z}_2$, $K = \mathbb{Z}_3$:

$$S = \text{Span}\{ 1, a^3, a^2 + a^4, a + a^5 \}$$

Examples of Schur Rings

Examples of Schur Rings:

- v) Let $H \triangleleft G$. Let S and T be S-rings over H and G/H , respectively. The partition which affords T over G/H can be inflated into a partition of G . The combination of the partitions of S and T afford an S-ring, $S \wedge T$, over G , called the **Wedge Product** of S and T .

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e.g. $G = \mathbb{Z}_6 = \langle a \rangle$, $H = \mathbb{Z}_3$, $G/H = \mathbb{Z}_2$:

$$S = \text{Span}\{ 1, a^2, a^4, a + a^3 + a^5 \}$$

These Schur rings are called the **Traditional Schur rings**.

Traditional Schur Rings

Traditional Schur rings are the best known Schur rings and the easiest to construct, but many finite groups are known to have non-traditional Schur rings.

Theorem - Shiu (1989)

There exists non-traditional Schur rings over dihedral groups.

Theorem - M (2014)

There exists non-traditional Schur rings over elementary abelian groups.

Traditional Schur Rings

Traditional Schur rings are the best known Schur rings and the easiest to construct, but many finite groups are known to have non-traditional Schur rings.

Theorem - Bastian, Brewer, Humphries, M, Thompson (2020)

There exists non-traditional Schur rings over free groups, free products, and HNN extensions.

Many attempts have been made to classify Schur rings over specific groups, but these attempts generally use external contributors to the group G , such as group actions and Schurian groups.

The traditional Schur rings are constructible entirely using internal properties of G .

Traditional Schur Rings

Theorem - Leung, Man (1996, 1998)

All Schur rings over finite cyclic groups are traditional.

(Trivial), (Automorphic), (Direct Product), (Wedge Product)

Traditional Schur Rings

Theorem - Bastian, Brewer, Humphries, M, Thompson (2020)

All Schur rings over the infinite cyclic group $\mathbb{Z}$ are traditional.

(Automorphic)

This proof relies on the Frobenius map and all non-identity elements having infinite order.

Theorem - Bastian, Brewer, Humphries, M, Thompson (2020)

All Schur rings over the virtually cyclic group $\mathbb{Z} \times \mathbb{Z}_2$ are traditional.

(Automorphic), (Wedge Product)

This proof relies again on the Frobenius map and all non-identity elements having infinite order except for one.

Traditional Schur Rings

Theorem - Bastian, Brewer, Humphries, M, Thompson (2018)

All Schur rings over a torsion-free locally cyclic group G are traditional.

(Automorphic)

This proof relies on the compatibility of Schur subrings over infinite cyclic subgroups.

Theorem - Bastian, Brewer, Humphries, M, Thompson (2018)

All Schur rings over the virtually locally cyclic group $G \times \mathbb{Z}_2$, where G is a torsion-free locally cyclic group, are traditional.

(Automorphic), (Wedge Product)

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Traditional Schur Rings

A prime p is **Fermat** if $p = 2^k + 1$. There are probably only five such primes.

A prime p is **safe** if $p = 2q + 1$ and q is itself prime (a Sophie Germain prime). There are probably infinitely many safe primes.

Theorem - Bastian, M (2019)

All Schur rings over the virtually cyclic group $\mathbb{Z} \times \mathbb{Z}_p$ are traditional, where p is a Fermat or safe prime.

(Automorphic), (Wedge Product)

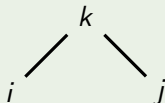
Directed Sets

Directed Sets

We say a partially ordered set I is **directed** if for all $i, j \in I$ there exists $k \in I$ such that

$$i, j \leq k,$$

that is, all pairs have a upper bound.

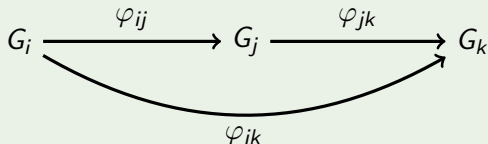


Direct Systems of Groups

Direct Systems of Groups

Let $\mathcal{G} = \{G_i \mid i \in I\}$ be set of groups G_i indexed by a directed set I . We say that $\mathcal{G}$ is a **direct system** of groups if there exists homomorphisms $\varphi_{ij} : G_i \rightarrow G_j$ such that:

- (1) φ_{ii} is the identity on G_i ;
- (2) $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ whenever $i \leq j \leq k$, that is, the following diagram is commutative:

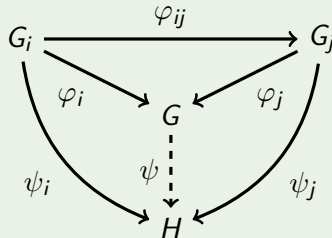


Direct Limits of Groups

The **direct limit** of a direct system $\mathcal{G}$, denoted $\varinjlim G_i$, is a group G equipped with homomorphisms $\varphi_i : G_i \rightarrow G$ such that:

- (1) $\varphi_i = \varphi_j \circ \varphi_{ij}$ whenever $i \leq j$;
- (2) if H is a **target** of G , that is, a group with homomorphisms $\psi_i : G_i \rightarrow H$ so that $\psi_i = \psi_j \circ \varphi_{ij}$, then there exists a unique homomorphism $\psi : G \rightarrow H$ so that $\psi_i = \psi \circ \varphi_i$.

In other words, the following diagram is commutative:



Direct Systems of Schur Rings

Direct Systems of Schur Rings

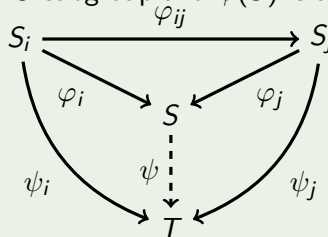
Let $\mathcal{G} = \{G_i \mid i \in I\}$ be a direct system of groups with maps $\varphi_{ij} : G_i \rightarrow G_j$. Let $\mathcal{S} = \{S_i \mid i \in I\}$ be a set of Schur rings S_i over groups G_i . We say that $\mathcal{S}$ is a **direct system** of Schur rings over $\mathcal{G}$ if:

- (1) $\ker \varphi_{ij}$ is an S_i -subgroup whenever $i \leq j$;
- (2) $\varphi_{ij}(S_i)$ is a Schur subring of S_j whenever $i \leq j$.

Direct Limits of Schur Rings

The **direct limit** of a direct system $\mathcal{S}$, denoted $\varinjlim S_i$, is a Schur ring S over $G = \varinjlim G_i$ such that:

- (1) $\ker \varphi_i$ is an S_i -subgroup and $\varphi_i(S_i)$ is a Schur subring of S for all i ;
- (2) if T is a Schur ring over a target of $\mathcal{G}$, say H , and if $\ker \psi_i$ is an S_i -subgroup and $\psi_i(S_i)$ is a Schur subring of T for all i , then $\ker \psi$ is an S -subgroup and $\psi(S)$ is a Schur subring of T .



Theorem - M (2019)

Let $\mathcal{S} = \{S_i\}_i$ be a direct system of Schur rings over $\mathcal{G} = \{G_i\}_i$.
Then the direct limit $\lim_{\rightarrow} S_i$ exists (and is unique) as a Schur ring
over $\lim_{\rightarrow} G_i$.

This allows us to define infinite **direct sums** and **direct products** of Schur rings, as these are limits of their finite sums/products.
We also get arbitrary unions of Schur rings.

Using direct limits, we can classify Schur rings over groups which are “almost” cyclic.

Theorem - M (2019)

If G is a torsion-free, locally cyclic group and all Schur rings over the virtually cyclic group $\mathbb{Z} \times \mathbb{Z}_n$ are traditional, then all Schur rings over $G \times \mathbb{Z}_n$ are traditional.

(Automorphic), (Direct Product), (Wedge Product)

Theorem - M (2019)

All Schur rings over a locally cyclic group G are traditional.

(Automorphic), (Direct Product), (Wedge Product)

Future Questions

What other virtually finite or locally finite groups might the Schur rings be classified over?

These techniques can be modified to the dual structure, inverse limits.

Conjecture

All Schur rings over pro-cyclic group, such as the p -adic integers, are traditional.

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What other virtually finite or locally finite groups might the Schur rings be classified over?

These techniques can be modified to the dual structure, **inverse limits**.

Conjecture

All Schur rings over pro-cyclic group, such as the p -adic integers, are traditional.

What other pro-finite groups might the Schur rings be classified over?

Thank you for listening!